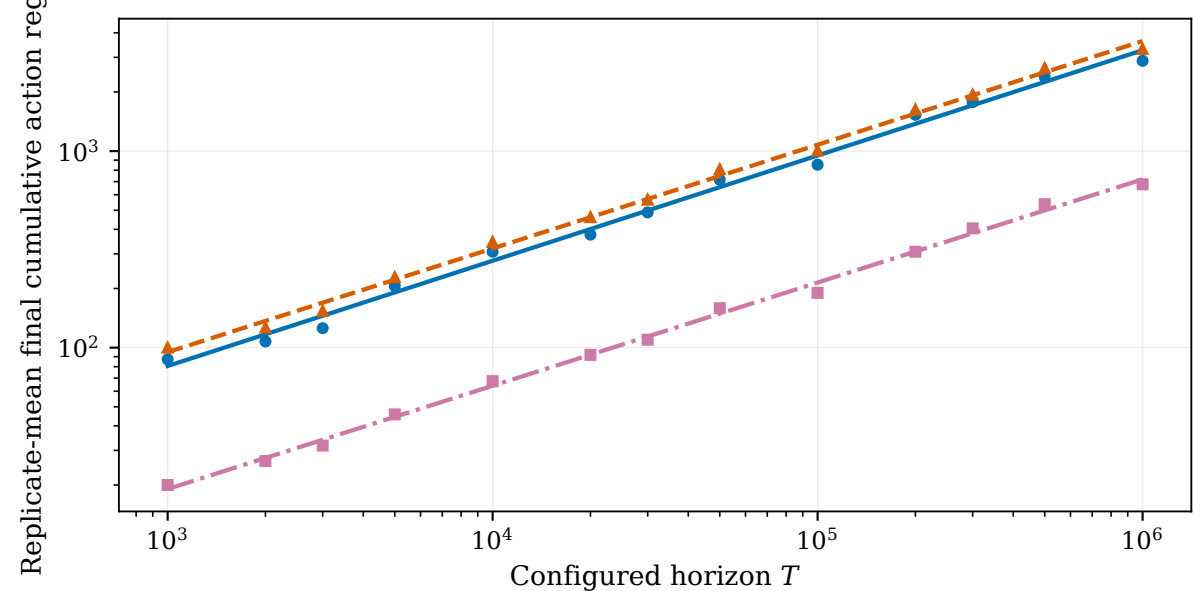


Experiment Information

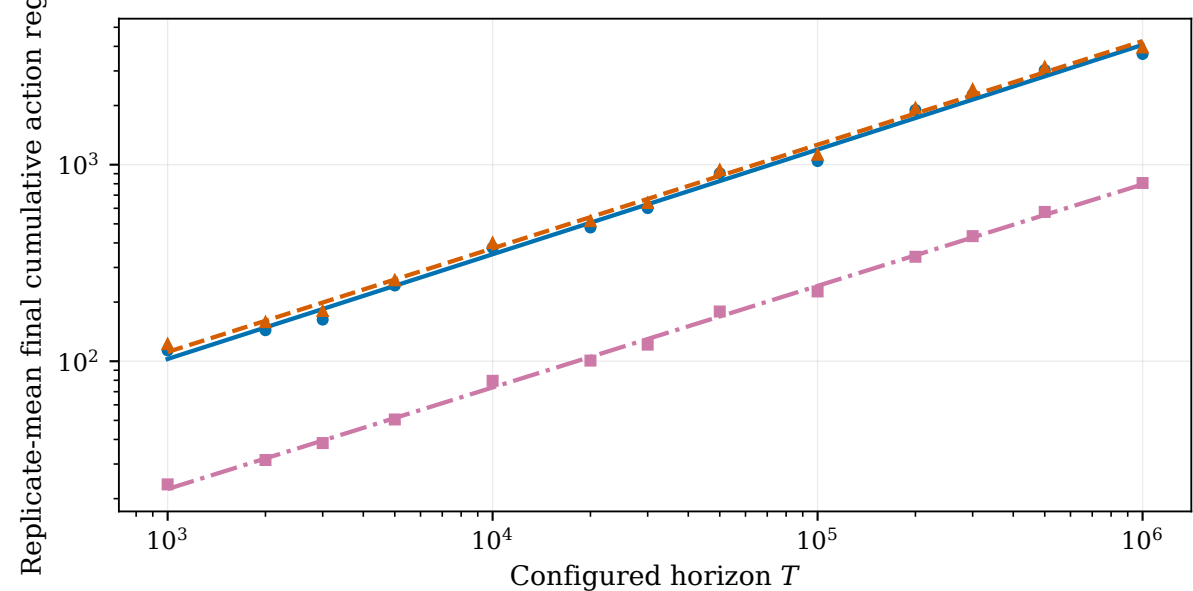
Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: EXP3
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.0028822, SD = 0.000932987, SE = 0.00026933
Internal R/T: mean = 0.000677917, SD = 0.000204923, SE = 5.91562e-05
Swap R/T: mean = 0.0033119, SD = 0.000741822, SE = 0.000214146
External: $\alpha = 0.536$, $c = 1.995$, $R^2 = 0.994$
Internal: $\alpha = 0.525$, $c = 0.5067$, $R^2 = 0.997$
Swap: $\alpha = 0.528$, $c = 2.479$, $R^2 = 0.997$



- External: $\alpha = 0.536, c = 1.995$
- Internal: $\alpha = 0.525, c = 0.5067$
- Swap: $\alpha = 0.528, c = 2.479$

Experiment Information

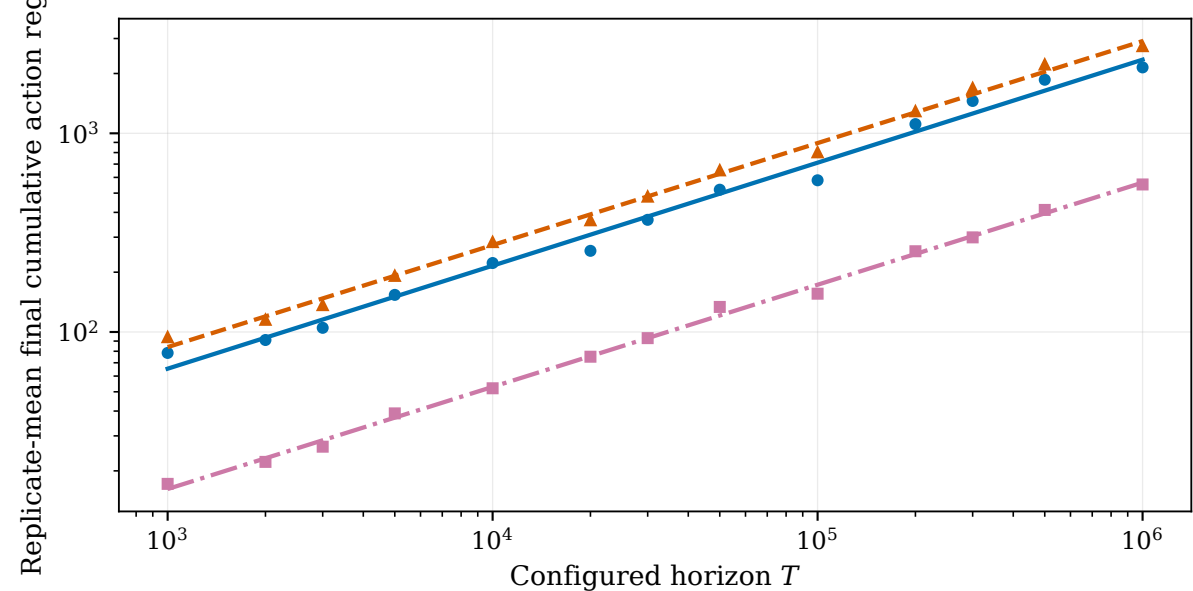
Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: BM-EXP3
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00365742, SD = 0.00150648, SE = 0.000434883
Internal R/T: mean = 0.000806575, SD = 0.000220401, SE = 6.36244e-05
Swap R/T: mean = 0.00394529, SD = 0.00136933, SE = 0.000395292
External: $\alpha = 0.533$, $c = 2.591$, $R^2 = 0.995$
Internal: $\alpha = 0.517$, $c = 0.6306$, $R^2 = 0.998$
Swap: $\alpha = 0.527$, $c = 2.921$, $R^2 = 0.996$



- External: $\alpha = 0.533, c = 2.591$
- Internal: $\alpha = 0.517, c = 0.6306$
- Swap: $\alpha = 0.527, c = 2.921$

Experiment Information

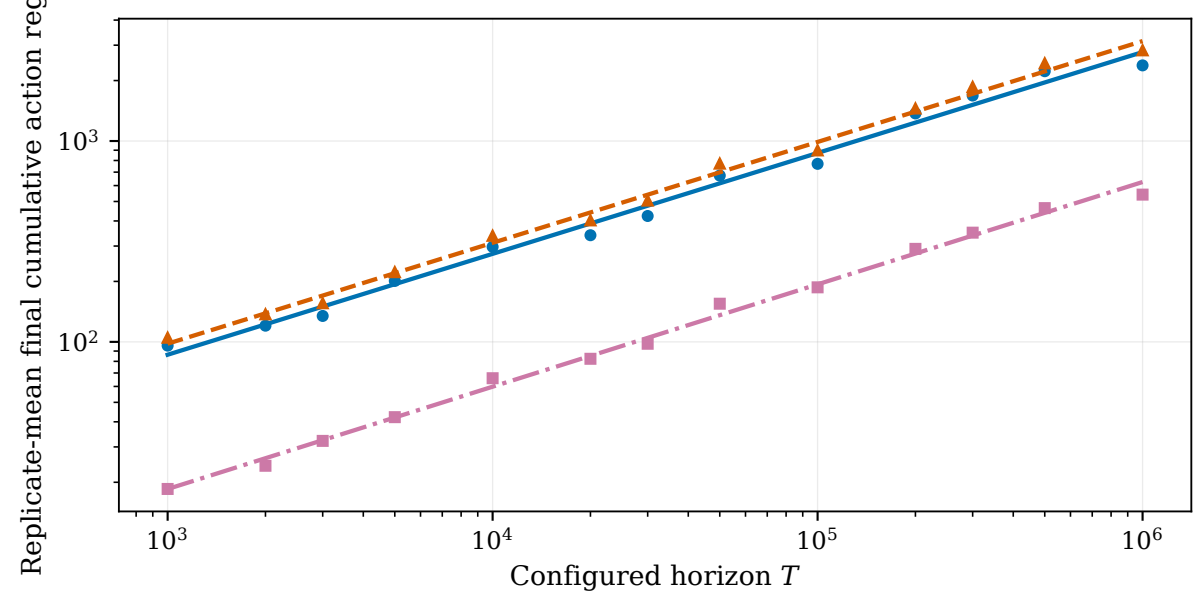
Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: EXP3-IX
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.002147, SD = 0.00113197, SE = 0.00032677
Internal R/T: mean = 0.000552792, SD = 0.00013908, SE = 4.01491e-05
Swap R/T: mean = 0.00274492, SD = 0.000825438, SE = 0.000238283
External: $\alpha = 0.518$, $c = 1.819$, $R^2 = 0.989$
Internal: $\alpha = 0.514$, $c = 0.4649$, $R^2 = 0.998$
Swap: $\alpha = 0.513$, $c = 2.42$, $R^2 = 0.996$



- External: $\alpha = 0.518, c = 1.819$
- Internal: $\alpha = 0.514, c = 0.4649$
- Swap: $\alpha = 0.513, c = 2.42$

Experiment Information

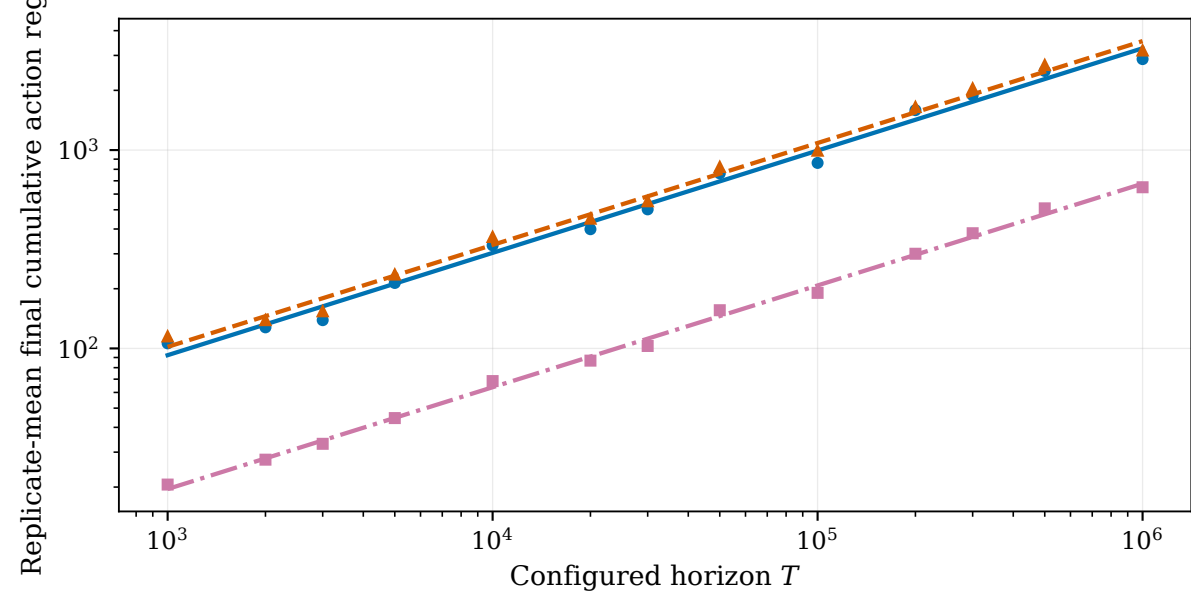
Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: Ito-Tsallis
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00237984, SD = 0.00127608, SE = 0.000368372
Internal R/T: mean = 0.00054095, SD = 0.000133307, SE = 3.84825e-05
Swap R/T: mean = 0.00282644, SD = 0.0010513, SE = 0.000303485
External: $\alpha = 0.502$, $c = 2.688$, $R^2 = 0.990$
Internal: $\alpha = 0.509$, $c = 0.5484$, $R^2 = 0.996$
Swap: $\alpha = 0.502$, $c = 3.051$, $R^2 = 0.995$



- External: $\alpha = 0.502$, $c = 2.688$
- Internal: $\alpha = 0.509$, $c = 0.5484$
- Swap: $\alpha = 0.502$, $c = 3.051$

Experiment Information

Figure: Horizon scaling
Compare: Horizons
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizons: 1,000, 2,000, 3,000, 5,000, 10,000, 20,000, 30,000, 50,000, 100,000, 200,000, 300,000, 500,000, 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
Aggregation: Mean final cumulative action regret across replicates
Fit model: $\log R_T = \alpha \log T + \log c$
Fit points: 13
Fit range: 1,000 to 1,000,000
Final endpoint at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External R/T: mean = 0.00287459, SD = 0.00137781, SE = 0.000397738
Internal R/T: mean = 0.000649375, SD = 0.000157445, SE = 4.54504e-05
Swap R/T: mean = 0.00317503, SD = 0.00120085, SE = 0.000346655
External: $\alpha = 0.516$, $c = 2.622$, $R^2 = 0.991$
Internal: $\alpha = 0.513$, $c = 0.5647$, $R^2 = 0.997$
Swap: $\alpha = 0.514$, $c = 2.926$, $R^2 = 0.994$



- External: $\alpha = 0.516, c = 2.622$
- Internal: $\alpha = 0.513, c = 0.5647$
- Swap: $\alpha = 0.514, c = 2.926$